

2023-24 - FIRST SEMESTER
FINANCIAL TIME SERIES (SFA01)
MASTER 2, PRINCIPAL SESSION

Written exam of January 29, 2024

Documents autorisés: feuille manuscrite recto-verso format A4.
Durée: 2 heures.

Authorized documents: handwritten double-sided sheet on A4 paper.
Duration: 2 hours.

1 Preamble

The following facts should be helpful.

Reminders on Gaussian distributions: Let $m \in \mathbb{R}^n$ and Σ be a symmetric non-negative semi-definite $\mathbb{R}^{n \times n}$ matrix. $\mathcal{N}(m, \Sigma)$ denotes the Gaussian distribution with mean m and covariance matrix Σ . It is defined on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and, if Σ is invertible (hence symmetric positive definite), it admits the density

$$x \mapsto (2\pi)^{-n/2} (\det(\Sigma))^{-1/2} e^{-\frac{1}{2}(x-m)^T \Sigma^{-1}(x-m)},$$

with respect to the Lebesgue measure on $\mathbb{R}^n$.

Reminders on linear prediction: Let $\vec{X}$ and $\vec{Y}$ be two L^2 random vectors valued in $\mathbb{C}^p$ and $\mathbb{C}^q$. Then if $\mathbb{E}[\vec{Y}] = 0$ and $\text{Cov}(\vec{Y})$ is invertible, we have

$$\text{proj}(\vec{X} \mid \{C \vec{Y} : C \in \mathbb{C}^{m \times m}\}) = \text{Cov}(\vec{X}, \vec{Y}) \text{Cov}(\vec{Y})^{-1} \vec{Y}.$$

Conditional distribution in the Gaussian case: If moreover $\begin{bmatrix} \vec{X}^T & \vec{Y}^T \end{bmatrix}^T$ is a Gaussian vector valued in $\mathbb{R}^{p+q}$, then the conditional distribution of $\vec{X}$ given $\vec{Y}$ is $\mathbb{P}^{\vec{X}|\vec{Y}}(\vec{Y}, \cdot) = \mathcal{N}(\hat{X}, \Sigma)$ where

$$\hat{X} = \mathbb{E}[\vec{X} \mid \vec{Y}] = \mathbb{E}[\vec{X}] + \text{proj}(\vec{X} \mid \{C(\vec{Y} - \mathbb{E}[\vec{Y}]) : C \in \mathbb{C}^{m \times m}\}),$$

and $\Sigma = \text{Cov}(\vec{X} - \hat{X})$.

Links between exercises The same model is used in Exercise 1 and Exercise 2. In Exercise 1, we use a state-space representation of the model. However this representation is not used in Exercise 2, which can be solved independently.

2 Exercises

Exercise 1. Let $m \geq 2$ and let us consider an m -dimensional time series $\vec{Y} = ((Y_t(k))_{1 \leq k \leq m})_{t \in \mathbb{Z}}$ valued in $\mathbb{R}^m$, defined by

$$\vec{Y}_t = \vec{Z}_t + \Theta \vec{Z}_{t-1}, \quad t \in \mathbb{Z},$$

where Θ is an $m \times m$ real matrix and $\vec{Z} \sim \text{WN}(0, \Sigma)$ is an m -dimensional centered white noise valued in $\mathbb{R}^m$ with covariance matrix Σ . We always assume that Σ is positive definite (hence invertible).

1. What is the nature of the process $\vec{Y}$? Compute $\mathbb{E}[\vec{Y}_t]$ and $\text{Cov}(\vec{Y}_t)$ for all $t \in \mathbb{Z}$. Show that $\text{Cov}(\vec{Y}_t)$ is positive definite.

2. Give the auto-covariance function and the spectral density matrix function of $\vec{Y}$.

Let $\vec{X}_t = \begin{bmatrix} \vec{Z}_t^T & \vec{Z}_{t-1}^T \end{bmatrix}^T$ for all $t \in \mathbb{Z}$.

3. Find a $(2m) \times (2m)$ matrix A such that $\vec{X}$ is solution of the equation

$$\vec{X}_t = A\vec{X}_{t-1} + \begin{bmatrix} \vec{Z}_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad t \in \mathbb{Z}.$$

4. Find a $m \times (2m)$ matrix B such that

$$\vec{Y}_t = B\vec{X}_t, \quad t \in \mathbb{Z}.$$

5. Justify that we can interpret the two previous equations for $t \geq 1$ as the state and observation equations of a dynamic linear model. Compute $\mathbb{E}[\vec{X}_0]$ and $\text{Cov}(\vec{X}_0)$.

We observe $\vec{Y}_1, \dots, \vec{Y}_n$ and define for all $s \geq 1$ the vector space

$$\mathcal{G}_{1:s}^{\vec{Y}} := \text{Span} \left(C \vec{Y}_u : 1 \leq u \leq s, C \in \mathbb{C}^{m \times m} \right).$$

We wish to compute

$$\begin{aligned} \vec{Y}_{s+1|s} &= \text{proj} \left(\vec{Y}_{s+1} \middle| \mathcal{G}_{1:s}^{\vec{Y}} \right), \\ \Gamma_{s+1|s} &= \text{Cov} \left(\vec{Y}_{s+1} - \vec{Y}_{s+1|s} \right). \end{aligned}$$

as well as the likelihood associated to the above model with unknown parameters (Θ, Σ) , assuming that $\vec{Z}$ is a Gaussian process. To this end, we introduce, for all $s, t \in \mathbb{N}^*$,

$$\begin{aligned} \vec{X}_{t|s} &= \text{proj} \left(\vec{X}_t \middle| \mathcal{G}_{1:s}^{\vec{Y}} \right), \\ \Sigma_{t|s} &= \text{Cov} \left(\vec{X}_t - \vec{X}_{t|s} \right). \end{aligned}$$

6. Determine $\text{Cov}(\vec{X}_1, \vec{Y}_1)$ and deduce $\vec{X}_{1|1}$.
7. Show that $\text{Cov}(\vec{X}_1 - \vec{X}_{1|1}, \vec{X}_1) = \text{Cov}(\vec{X}_1 - \vec{X}_{1|1}, \vec{X}_1)$ and deduce $\Sigma_{1|1}$.
8. Let $t \geq 2$. Use the state equation to deduce $\vec{X}_{t|t-1}$ and $\Sigma_{t|t-1}$ from $\vec{X}_{t-1|t-1}$ and $\Sigma_{t-1|t-1}$.
9. Express $\vec{Y}_{t|t-1}$ and $\Gamma_{t|t-1}$ using $\vec{X}_{t|t-1}$ and $\Sigma_{t|t-1}$.
10. Show that we have
$$\mathcal{G}_{1:t}^{\vec{Y}} = \mathcal{G}_{1:(t-1)}^{\vec{Y}} \oplus^\perp \left\{ C \left(\vec{Y}_t - \vec{Y}_{t|t-1} \right) : C \in \mathbb{C}^{m \times m} \right\}.$$
11. Deduce an expression of $\vec{X}_{t|t}$ using $\vec{X}_{t|t-1}$, $\vec{Y}_t - \vec{Y}_{t|t-1}$, $\Gamma_{t|t-1}$ and $\Sigma_{t|t-1}$.
12. Compute $\Sigma_{t|t}$ using $\Gamma_{t|t-1}$ and $\Sigma_{t|t-1}$.
13. Deduce an iterative algorithm to compute $\vec{Y}_{t|t-1}$ and $\Gamma_{t|t-1}$ for all $t = 2, \dots, n+1$.
14. Explain how this algorithm can be used to compute the log-likelihood associated to the above model for $\vec{Y}_1, \dots, \vec{Y}_n$ with unknown parameters (Θ, Σ) .

Exercise 2. We consider the same time series $\vec{Y}$ and $\vec{Z}$ as in Exercise 1. We now wish to examine the instantaneous Granger causality for $\vec{Y}$.

15. Provide a condition on Θ implying that the linear past of $\vec{Y}$ coincides with the linear past of $\vec{Z}$, that is, $\mathcal{H}_t^{\vec{Y}} = \mathcal{H}_t^{\vec{Z}}$ for all $t \in \mathbb{Z}$, where
$$\mathcal{H}_t^{\vec{Y}} = \overline{\text{Span}} \left(v^H \vec{Y}_s : s \leq t, v \in \mathbb{C}^m \right),$$

$$\mathcal{H}_t^{\vec{Z}} = \overline{\text{Span}} \left(v^H \vec{Z}_s : s \leq t, v \in \mathbb{C}^m \right).$$

From now on, we assume that $\mathcal{H}_t^{\vec{Y}} = \mathcal{H}_t^{\vec{Z}}$ for all $t \in \mathbb{Z}$.

16. Show that, for all $k = 1, \dots, m$,
$$\text{proj} \left(\vec{Y}_t(k) \middle| \mathcal{H}_{t-1}^{\vec{Y}} \right) = \left[\Theta \vec{Z}_{t-1} \right] (k).$$

We now take two integers $k \neq \ell$ between 1 and m .

17. Show that $\mathcal{H}_{t-1}^{\vec{Y}} + \mathcal{H}_t^{\vec{Y}^{(\ell)}} = \mathcal{H}_{t-1}^{\vec{Y}} + \mathcal{H}_t^{\vec{Z}^{(\ell)}}$, where
$$\mathcal{H}_t^{\vec{Y}^{(\ell)}} = \overline{\text{Span}} \left(\vec{Y}_s(\ell) : s \leq t \right),$$

$$\mathcal{H}_t^{\vec{Z}^{(\ell)}} = \overline{\text{Span}} \left(\vec{Z}_s(\ell) : s \leq t \right).$$

18. Deduce an expression of $\text{proj} \left(\vec{Y}_t(k) \middle| \mathcal{H}_{t-1}^{\vec{Y}} + \mathcal{H}_t^{\vec{Y}^{(\ell)}} \right)$ involving $\vec{Z}_{t-1}$ and $\vec{Z}_t$.
19. Provide a necessary and sufficient condition on Σ to have that $\vec{Y}^{(\ell)}$ instantaneously $\vec{Y}$ -Granger causes $\vec{Y}^{(k)}$.